

Nhat Tran

FabLab Technician, Eastern International University

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SUMMARY

I work on autonomous robots and drones across the full stack — from PCB design and low-level firmware (SPI, I2C, CAN bus, Ethernet, Modbus on RTOS) to Visual SLAM, computer vision and the ROS 2 navigation stack. I also run weekly workshops on robotics and embedded systems.

EDUCATION

Control Engineering and Automation

2015 – 2022

Eastern International University (EIU), Vietnam

- Completed full program coursework
- **Focus:** control systems, embedded systems, robotics
- **Relevant coursework:** Automatic Control, Microcontrollers, Power Electronics, PLC & Industrial Automation, Digital Signal Processing

EXPERIENCE

FabLab Technician

Oct 2018 – Present

Eastern International University (EIU)

- Develop autonomous robot and autonomous drone platforms, working across the full stack from PCB design and low-level firmware to high-level autonomy.
- **Technical Lead** (Jun–Nov 2026), joint EIU–ARTC (Singapore) project [VDA5050 Support for Open-RMF](#): ROS 2 fleet adapters, an MQTT client adapter with a Nav2 bridge, and a fleet dashboard for a mixed AMR / TurtleBot3 fleet.
- **Autonomy & perception:** Visual SLAM, computer vision and ROS 2 navigation — custom Nav2 planner and controller plugins, ros2_control hardware interfaces.
- **Embedded:** PCB design for robot controller boards in Altium Designer, and low-level firmware on RTOS — SPI, I2C, UART, CAN bus, Ethernet and Modbus.
- Organize and teach a weekly workshop series for students on robotics, ROS 2, drones and embedded systems.
- Mentor student teams in annual robotics competitions — **2nd and 3rd Prize at AUTO RACE 2025** with two teams.
- Supervise student research papers — **Best Paper Award (Student Session)** at the EIUSC International Conference 2025.

PUBLICATIONS

- [1] Hoang Anh Nguyen Trong, Thuy Dao Le Ngoc, Toan Pham Dang Huu, Uyen Vo Pham Mai, **Duy Nhat Tran**. *Data-Driven Feedforward-Assisted PI Speed Control with Automatic Position Holding for Geared DC Motors*. Under review at EIUSC International Conference 2026.
- [2] Duong Tai Au, Cong Danh Huynh, **Duy Nhat Tran**. *Sim-to-Real Transfer for End-to-End Autonomous Racing via Imitation Learning and Control Barrier Function Safety Filtering*. Under review at EIUSC International Conference 2026.
- [3] Cong Danh Huynh, Nhu Y Pham, Hoang Dung Bui, Duong Tai Au, **Duy Nhat Tran**. *Developing An Autonomous Emergency Braking System-Based Kalman Filtering For An Autonomous Racing Car*. EIUSC International Conference 2025 — Student Session, Ho Chi Minh City. [\[Best Paper Award \(Student Session\)\]](#)

SKILLS

Embedded & Hardware	PCB Design (Altium Designer), RTOS, SPI, I2C, UART, CAN bus, Ethernet, Modbus
Perception & AI	PyTorch, OpenCV, TensorRT, ONNX, YOLO, PCL
Autonomy	ROS 2, Nav2, ros2_control, Visual SLAM, Open-RMF, VDA5050, MQTT, Autonomous Robots, Autonomous Drones
Programming	C / C++, Python, CUDA, Bash
Tools	Git, Docker, Linux, NVIDIA Jetson
Languages	Vietnamese (native), English (professional)

ACHIEVEMENTS

Best Paper Award (Student Session) — EIUSC International Conference 2025

Dec 2025

As research mentor and co-author

- Paper: *Developing An Autonomous Emergency Braking System-Based Kalman Filtering For An Autonomous Racing Car*
- Oral presentation at EIUSC 2025, Ho Chi Minh City, 3 December 2025.
- [Experiment video on YouTube](#)

2nd Prize & 3rd Prize — AUTO RACE 2025

2025

As mentor of two EIU student teams

- Southern Vietnam autonomous vehicle competition, hosted by FPT University Can Tho.
- [Read the story on EIU News](#)

ACTIVITIES

Mentor — Team EIU-Spark (ongoing)

2026 – Present (competition: October 2026)

Ho Chi Minh City Youth UAV-Drone Creativity and Control Competition 2026

- Project **EIU-Spark Delivery UAV**: semi-autonomous delivery drone that flies to a pickup station, identifies the parcel by ArUco marker, picks it up and delivers it to the drop-off point.
- *Cuộc thi Sáng tạo và Điều khiển UAV-Drone dành cho Thanh thiếu niên TP. Hồ Chí Minh năm 2026*

Weekly Robotics Workshop Series — Organizer & Instructor

Jan 2025 – Present

EIU FabLab

- Hands-on sessions for students on robotics, ROS 2, drones and embedded systems.
- Workshop *STM32 And Two-Wheeled Self-Balancing Robot*: C, STM32 peripherals, IMU calibration, attitude estimation (EKF) and LQR balancing control — [lecture recordings](#).
- Workshop *F1TENTH Autonomous Racing*: Ubuntu, Python, ROS 2, AEB, PID wall following, SLAM Toolbox mapping, AMCL localization, Pure Pursuit and RRT / RRT* obstacle avoidance — [lecture recordings](#).

Student Research Mentor

EIU FabLab

- Guide student teams in writing and presenting scientific research papers.
- EIUSC 2026 (under review, co-author): *Data-Driven Feedforward-Assisted PI Speed Control with Automatic Position Holding for Geared DC Motors* — [demo video](#)
- EIUSC 2026 (under review, supervising author): *Sim-to-Real Transfer for End-to-End Autonomous Racing via Imitation Learning and Control Barrier Function Safety Filtering* — [demo video](#)
- EIUSC International Conference 2025: our student paper received the **Best Paper Award (Student Session)** — [experiment video](#).

Student Competition Mentor

EIU FabLab

- Mentor student teams for annual robotics competitions.
- AUTO RACE 2025 (Southern Vietnam autonomous vehicle competition, FPT University Can Tho): two teams won **2nd Prize** and **3rd Prize** — [read on EIU News](#).